# KYLE DAWSON, PHD

## PhD Applied Scientist | Mathematical Optimization, Machine Learning & Generative AI

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## EXPERIENCE

### Lead Data Scientist — Energy

**Centerpoint Energy**

Apr 2026 – Present Houston, TX

- lead a new applied-science team, setting technical direction, mathematical-modeling and evaluation standards, and mentoring scientists and engineers across machine-learning and optimization initiatives
- designed and productionized a large-scale budget-constrained optimization model (mixed-integer linear programming via CBC with greedy heuristics, LightGBM risk scoring) that allocates a fixed operational budget to maximize grid reliability (SAIFI) per dollar — translating an ambiguous business problem into a rigorous OR formulation deployed on GCP
- built a deployed multi-LLM / generative-AI application (Gemini on Vertex AI, Cohere North) on Cloud Run with a retrieval benchmark scoring citation accuracy against a ground-truth set, to evaluate foundation models for document Q&A and image analysis
- ship production-grade Python systems (Docker, CI/CD, BigQuery) that take models from research prototype through simulation and experimentation to reliable, large-scale production

### Data Scientist → Senior Data Scientist — Climate Tech

**SeekOps Inc.**

Mar 2022 – Apr 2026 Austin, TX

- built and productionized machine-learning models to forecast survey quality, detect anomalies, and constrain uncertainty in gas-emission estimates, delivered through customer-facing analytics platforms (Plotly Dash on Azure) securing contracts of $50–150k
- implemented Bayesian uncertainty propagation and physics-based simulation models to generate decision-grade metrics used in regulatory and enterprise reporting across 10+ countries
- invented novel measurement and quantification methodologies, presenting results to executives, customers, and international peer-reviewed conferences

### Data Scientist — Ag Tech

**Bayer CropScience**

Jan 2020 – Mar 2022 St. Louis, MO

- applied machine learning and statistical modeling to large-scale UAV-derived imagery for digital phenotyping and crop-performance assessment
- trained and evaluated deep-learning object-detection models (Detectron2) for classification from high-resolution aerial imagery
- built end-to-end ML data workflows combining relational databases, geospatial data, and AWS S3 into machine-learning-ready datasets serving stakeholder decision support

### Postdoctoral Fellow

**NASA Langley Research Center**

Feb 2018 – Jan 2020 Hampton, VA

- developed uncertainty-aware retrieval algorithms from remote-sensing observations and delivered peer-reviewed results, prototyping and experimenting with models against field data

## SELECTED PROJECT

### BESS Day-Ahead Arbitrage Optimizer

**Self-directed proof of concept**

## EXECUTIVE SUMMARY

PhD applied scientist with 8+ years inventing, experimenting with, and productionizing machine-learning, optimization, and simulation systems for scientifically complex, large-scale problems. Combines mathematical optimization (MILP/LP), machine learning, and generative AI to turn ambiguous business problems into rigorous models and shipped software. Peer-reviewed researcher who carries solutions from prototype and simulation through experimentation into reliable cloud production, and communicates results to technical and business stakeholders.

## CORE COMPETENCIES

Mathematical Optimization · Operations Research · Research & Invention · Simulation & Experimentation · Research-to-Production · Model Evaluation & Validation · Technical Leadership · Mentoring & Team Building · Cross-Functional Collaboration · Stakeholder Communication

## DOMAIN EXPERTISE

Mathematical Optimization (MILP/LP) · Machine Learning · Deep Learning · Forecasting & Time Series · Generative AI & LLMs · Anomaly Detection · Bayesian Inference · Uncertainty Quantification · Physics-Based Simulation · Computer Vision · Geospatial & Remote Sensing

## TOOLS & TECHNOLOGIES

Python · PuLP / CBC · PyTorch · Scikit-Learn · LightGBM · XGBoost · Detectron2 · NumPy · Pandas · SQL · GCP BigQuery · Vertex AI · AWS · Azure · Docker · GPU / CUDA · MLOps / CI-CD · Git · Plotly · Anthropic Claude

## AWARDS

### NASA Postdoctoral Program Fellow

**NASA Langley Research Center**

Oct 2018 Hampton, VA

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### Langley Aerospace Research Scholar

**NASA Langley Research Center**

Jun 2013 Hampton, VA

## PUBLICATIONS

Google Scholar Profile

2025

A PuLP/CBC linear program, fed by ERCOT market API data, that schedules day-ahead battery charge/discharge arbitrage against nodal settlement-point (LMP) prices under state-of-charge and power constraints.

## EDUCATION

### PhD, Atmospheric Science

**North Carolina State University**

Graduated 2017 Raleigh, NC

Contributions to the Understanding of Aerosol Microphysics Towards Improving the Assessment of Climate Radiative Forcing